Lingbo Tong

CONTACT INFORMATION

1071 Educational Sciences
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RESEARCH INTERESTS

AI-Enhanced Psychometric Methods, Natural Language Processing, Deep Learning, Computational Social Science.

ACADEMIC APPOINTMENTS

University of Wisconsin-Madison, Madison, WI
Quantitative Methods, Department of Educational Psychology
Assistant Professor — 2026–
Anna Julia Cooper Research Fellow — 2025–2026

EDUCATION

University of Notre Dame, Notre Dame, IN
Joint Ph.D. in Quantitative Psychology and Computer Science and Engineering — 2025
Advisors: Prof. Zhiyong (Johnny) Zhang and Prof. Meng Jiang
M.S. in Computer Science and Engineering — 2024

Sichuan University, Chengdu, China
B.Eng. in Computer Science and Technology — 2020

PUBLICATIONS

Gao, Z., **Tong, L.**, & Zhang, Z. (2026). Detecting and Evaluating Bias in Large Language Models: Concepts, Methods, and Challenges. Journal of Behavioral Data Science, 6(1), 1-68. [paper]

Tong, L., & Zhang, Z. (2025). Neural Network Analysis of Psychological Data: A Step-by-Step Guide. *Multivariate Behavioral Research.* [paper]

Tong, L. (2025). Nonlinear Structural Equation Modeling with Text Data. *Dissertation*. [paper]

Wan R.*, **Tong, L.***, Knearem T., Li T., Huang K., & Wu Q. (2024). Hashtag Re-appropriation For Audience Control on Recommendation-driven Social Media Xiaohongshu. *The ACM Conference on Human Factors in Computing Systems (CHI)*. Honorable Mention for Best Paper (Top 5%). [paper]

Tong, L., Qu, W., & Zhang, Z. (2024). Comparison of the K1 Rule, Parallel Analysis, and the Bass-Ackward Method on Identifying the Number of Factors in Factor Analysis. *Fudan Journal of the Humanities and Social Sciences,* 1-28. [paper]

Lu, Y., **Tong, L.**, & Cheng, Y. (2024). Advanced Knowledge Tracing for Intelligent Tutoring Systems: Incorporating Process Data and Curricula Information via an Attention-Based Framework for Accuracy and Interpretability. *Journal of Educational Data Mining.* [paper]

Tong, L., Liu, Q., Yu, W., Yu, M., Zhang, Z., & Jiang, M. (2023). Improving mental health support response generation with event-based knowledge graph. *Workshop on Knowledge-Augmented Methods for NLP (KnowledgeNLP) at AAAI Conference on Artificial Intelligence (AAAI).* [paper]

Jiang M., Dang H., & **Tong L.** (2023). A Quantitative Review on Language Model Efficiency Research. Large Language Model. *Symposium at International Joint Conference on Artificial Intelligence (IJCAI).* [paper]

Wan, R., & **Tong, L.** (2023). Digital and Historical Exclusivity in Feminine Linguistics: From Nüshu to Xiaohongshu. *WiNLP workshop at Conference on Empirical Methods in Natural Language Processing (EMNLP).* [paper]

Yu, M., Yu, W., **Tong, L.**, & Jiang, M. (2022). Scientific Comparative Argument Generation. *Document Intelligence Workshop (DI) at ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD).* [paper]

Kuebler, J., **Tong, L.**, & Jiang, M. (2021). Multi-Round Parsing-Based Multiword Rules for Scientific Knowledge Extraction. *IEEE International Conference on Big Knowledge (ICBK)* (pp. 331-338). IEEE. [paper]

Chen, H., Jin S., Lin W., Zhu Z., **Tong L.**, Liu Y., ... & Sun M. (2021). Quantitative analysis on the Communication of COVID-19 Related Social Media Rumors. (In Chinese). *Journal of Computer Research and Development*, 58(7), 1366-1384. [paper]

Presentations and Posters

Kim, J.S., **Tong, L.**, & Chambers, M. (2026). Text Integration Methodology: A Framework for Constructing Hybrid Variables. *International Meeting of the Psychometric Society*, 21–24 July, Seoul, Republic of Korea.

Chen, W., **Tong, L.**, & Gant, A. (2026). Uncovering Experiences and Perceptions of Nonconsensual Condom Deception (NCCD) Through Social Media Text-Mining. *Midwest Nursing Research Society (MNRS) Annual Research Conference*, 30 March–2 April, St. Louis, MO, USA.

Tong, L. & Jiang, M. (2024). Large Language Models in Mental Health Support. Invited talk for *Summer Research Experience for Teacher's (RET) Program*, 19 June, Notre Dame, USA.

Tong, L., Zhang, Z., Jiang M., & Li, J. (2023). Permutation test of Importance-Weighted Autoencoder for Factor Analysis. *Symposium at Lucy Family Institute for Data & Society*, 7 September, Notre Dame, USA.

Tong, L. & Zhang, Z. (2023). Evaluation of the Bass-Ackward Method for Identifying the Number of Factors. *Annual Meeting of the International Society for Data Science and Analytics*, 4–6 July, Shanghai, China.

Tong, L., Liu, Q., Yu, W., Yu, M., Zhang, Z., & Jiang, M. (2023). MHKG: Improving mental health support response generation with event-based knowledge graph. *KnowledgeNLP at AAAI*, 7–14 February, Washington DC, USA.

Zhang, J., Chen, H., & **Tong, L.** (2022). Structured Gender Bias: How is Gender Cognition Changed and Affected by Digital Communities in China? *Annual Conference of International Association for Media and Communication Research Conference (IAMCR)*, 11–15 July, Beijing, China (online).

Tong, L., Zhang, Z., Jiang M., & Li J. (2022). Estimating Structural Equation Models with Neural Networks. *Symposium at Lucy Family Institute for Data & Society*, 28 September, Notre Dame, USA.

Software Development

Tong L., & Zhang, Z. (2024). TextSEM: An R Package for Structural Equation Modeling with Text Data. [code]

Tong L., & Zhang, Z. (2022). An Online Tool for Bass-Ackward Factor Analysis. [online app]

Xu, J., **Tong L.**, & Zhang, Z. (2020). Webnetvis: An Online Tool for Network Visualization. [online app]

Honors and Awards

Center for Research Computing (CRC) Graduate Award ($1,000) 2025
University of Notre Dame

ISLA Graduate Student Research Award ($4,000) 2024
University of Notre Dame

First place, 2023 EDM Cup 2023
International Educational Data Mining Society (IEDMS)

Summer Institute of Computational Social Science (SICSS) Scholarship 2023
NYU Shanghai Center for Applied Social and Economic Research (CASER)

Most Innovative Project Award 2019
Google AI ML Winter Camp, Beijing

First Prize Academic Scholarship 2019
Sichuan University

Outstanding Undergraduate Student Award 2017, 2018, 2019
Sichuan University

National Innovation Training Grant for Undergraduates 2017, 2018
Sichuan University

Teaching Experience

Co-Instructor, Grad Seminar: Quantitative Study (PSY 63199) Spring 2025, Fall 2024
Department of Psychology, University of Notre Dame

Teaching Assistant/Lab Instructor, Methods for Behavioral Sciences (PSY 30160) Fall 2023, Spring 2022
Department of Psychology, University of Notre Dame

Mentored Students

- Ruiting Shen, UW–Madison Ph.D. student in quantitative methods 2026–
- Sevantee Basu, UW–Madison Ph.D. student in quantitative methods 2026–
- Xuanjia Qiao, ND iSURE undergraduate student in computer science 2025
- Noah Crockett, ND undergraduate student in psychology 2024
- Anna Krush, ND master's student in applied and computational mathematics and statistics 2023–2024
- Yue Wan, ND exchange undergraduate student in data science 2023–2024
- Ishita Masetty, Penn High School student 2022
- Qi Liu, ND iSURE undergraduate student in computer science 2022
- Longqing Chen, ND undergraduate student in computer science 2021

Professional Leadership & Service

Workshop Co-Organizer:

- The Second Workshop on Knowledge Augmented Methods for NLP at ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 6–10 August, 2023, Long Beach, CA, USA. [website]
- Workshops at the Annual Meeting of the International Society for Data Science and Analytics, 21–24 July, 2024, Vienna, Austria. [website]

Manuscript Reviewer:

- Journal of Behavioral Data Science (JBDS)
- Methods in Psychology
- Journal of Family Psychology
- ACM Conference on Human Factors in Computing Systems (CHI)
- Annual Meeting of the Association for Computational Linguistics (ACL)
- Conference on Empirical Methods in Natural Language Processing (EMNLP)
- China National Conference on Computational Linguistics (CCL)

University Service:

- Departmental DEI Committee Representative 2023 – 2024
- Mentor for STEMentorship Program, Association with Women in Science 2023 – 2024
- Departmental Faculty Meeting Representative 2022 – 2023

TECHNICAL SKILLS

- Programming Languages: Python, R, JavaScript, SQL, HTML and others
- Quantitative Methods: Structural Equation Modeling, Multivariate Analysis, Longitudinal Data Analysis, Bayesian Statistics, Item Response Theory
- Machine Learning Topics: Explainable Machine Learning, Generative Language Models, Empathetic AI, Human-AI Collaboration